

Cybersecurity Incident Report: Network Traffic Analysis

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During the DNS lookup process, the UDP protocol was used to contact the DNS server in order to obtain the IP address for yummyrecipesforme.com. An ICMP error message was returned, indicating a failure to reach the DNS server.

Root Cause Analysis

The first two lines of each log entry show the UDP request sent from the browser to the DNS server. The third and fourth lines display the ICMP error response stating “UDP port 53 unreachable.” Since port 53 is used for DNS traffic, this indicates a DNS server issue.

Additional evidence of a DNS-related problem includes the plus sign (+) following the query ID 35084 and the “A?” flag, both of which indicate DNS query activity. Based on the ICMP error and DNS flags, it is highly likely that the DNS server is not responding to requests.

Incident Summary

The incident occurred today at 1:24 p.m., when customers reported receiving a “destination port unreachable” message while attempting to access yummyrecipesforme.com. The cybersecurity team providing IT services is currently investigating the issue to restore customer access to the website.

Technical Investigation

As part of the investigation, packet sniffing was performed using tcpdump. Analysis of the log files showed that DNS port 53 was unreachable. The next step is to determine whether the DNS server is down or if traffic to port 53 is being blocked by the firewall.

Possible causes include a Denial of Service (DoS) attack or a DNS server misconfiguration.

Appendix: Raw tcpdump Logs

13:24:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A? yummyrecipesforme.com. (24)

13:24:36.098564 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 254

13:26:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A? yummyrecipesforme.com. (24)

13:27:15.934126 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 320

13:28:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A? yummyrecipesforme.com. (24)

13:28:50.022967 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 150